
Flow Past a Cylinder

3D Simulation Analysis Report

Simulation Case	domain_D8X600Y120/uniform
Reynolds Number	$Re = 4 \times 10^6$
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1. Case Overview

1.1 Simulation Parameters

The following table summarizes the key parameters of the simulation case.

Parameter	Value
Domain X	600 m
Domain Y	120 m
Domain H (spanwise)	40 m
Cylinder Diameter D	8 m
Freestream Velocity U	0.5 m/s (from the 0 file)
Reynolds Number	4×10^6
Fluid Density	Uniform ambient
Solver	pimpleFoam (from constant/controlDict)
Turbulence Model	Spalart-Allmaras DDES (from constant/turbulenceProperties)
Time Range Simulated	0 to 1200 seconds (from system/controlDict)
Simulation Type	LES
Number of Processors	256

1.2 Mesh Overview

The simulation was run using pimpleFoam, a transient solver for turbulent incompressible flow, for a total of 1200 seconds with a time step of 0.01 seconds. The mesh contains around 43 million cells in total, split across 256 processor ($n_{\text{Cells}} = 168,126$ per processor, thus $256 * 168126 =$ around 43 million) in an $8 * 8 * 4$ arrangement in the X, Y, Z directions. The fluid is water with kinematic viscosity of $1.0 \times 10^{-6} \text{ m}^2/\text{s}$.

Parameter	Value	Source
Cells per processor	168,126	processor0/constant/polyMesh/owner → nCells:168126
Number of processors	256	system/decomposeParDict → numberOfSubdomains 256;
Processor arrangement	$8 \times 8 \times 4$	system/decomposeParDict → n (8 8 4);
Kinematic viscosity	$1.0 \times 10^{-6} \text{ m}^2/\text{s}$	constant/transportProperties → nu 1.0e-06;
Solver	pimpleFoam	system/controlDict → application pimpleFoam;
End time	1200 s	system/controlDict → endTime 1200;
Time step	0.01 s	system/controlDict → deltaT 0.01;

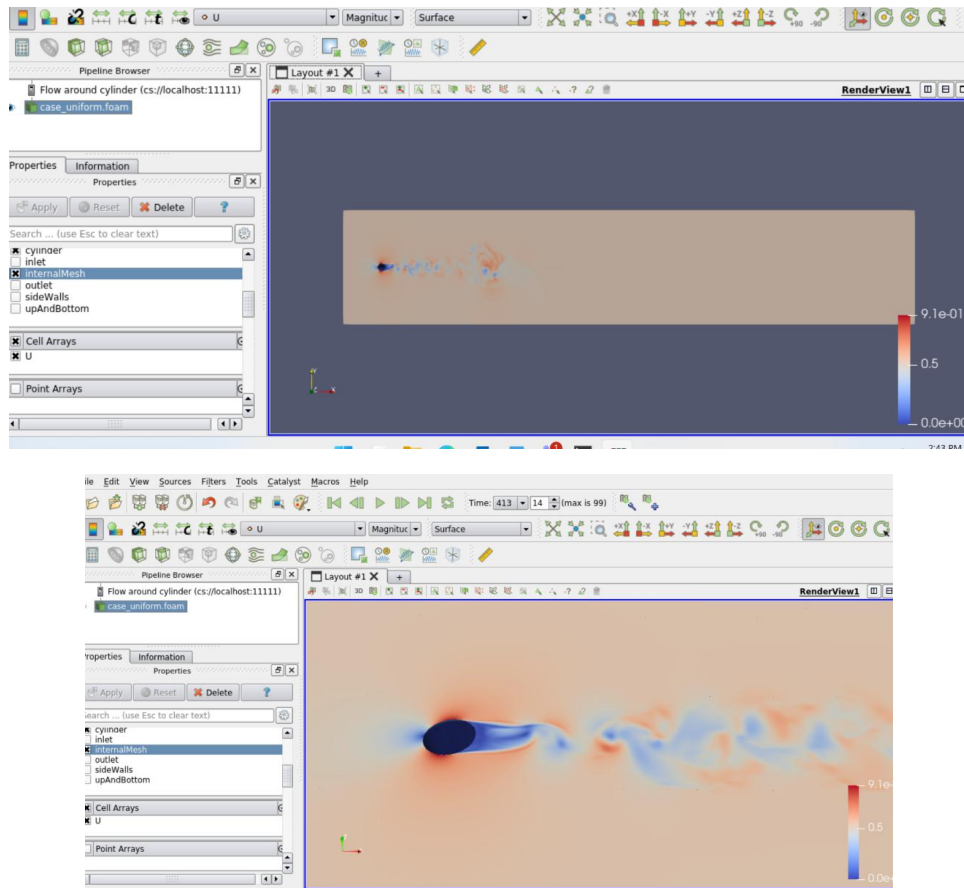


Figure 1.2 — ParaView views of the cylinder case: full domain (top) and zoomed wake region (bottom).

2. Post-Processing Methodology

2.1 Software and Tools Used

- ParaView 5.7.0
- Python 3.8.10 with NumPy and Matplotlib for FFT and plotting
- OpenFOAM (simulation was pre-run)

2.2 Approach

The simulation case was left in decomposed parallel form across 256 processor folders and was not reconstructed prior to post-processing. The case_uniform.foam file was opened in ParaView through MobaXTerm using the following method:

Terminal 1: Launch the parallel ParaView server

```
1 mpirun -np 16 pvserver
```

Terminal 2: Launch ParaView and connect

```
1 paraview
```

Then go to File → Connect, confirm the correct port number, and connect. Open case_uniform.foam, set Case Type to Decomposed Case, and press Apply.

3. Results

3.1 Mean Drag Coefficient (Cd)

Physical meaning: A dimensionless measure of the drag force acting on the body, normalized by dynamic pressure and a reference area. For a cylinder in cross-flow, it represents the average resistance force due to pressure and viscous effects.

Formula:

$$C_D(t) = \frac{F_D(t)}{\frac{1}{2}\rho U_\infty^2 D}$$

where F_D is the instantaneous drag force, ρ is fluid density, U_∞ is free-stream velocity, and D is cylinder diameter.

Mean value:

$$\overline{C_D} = \frac{1}{T} \int_0^T C_D(t) dt$$

From simulation: Compute the instantaneous drag force on the cylinder surface by integrating surface pressure and viscous stresses in the streamwise direction. Then, average over several vortex shedding cycles.

Figure 3.1a — Physical meaning, formula, and mean-value definition of the drag coefficient.

Variable	Value	Source
Reference Length D	8 m	Domain name
Spanwise Length L	40 m	Domain H
Freestream Velocity U	0.5 m/s	0/U (internalField / inlet value)
Reference Area D × L	320 m ²	Calculated
Integrated Drag Force	14.968 N	ParaView → Integrate Variables
Time-Averaged Cd	0.374202	Plot Data Over Time -> Excel Sheet = AVERAGE(C1:C101)
Reference	Link to Plot Over Line CSV file	

The time-averaged drag coefficient obtained from the simulation was $C_d = 0.374$. This value falls within the expected range for a circular cylinder in the supercritical/post-critical Reynolds number regime ($Re \approx 4 \times 10^6$), where reported drag coefficients are typically between 0.2 and 0.5. The use of the reference area $D \times L = 320 \text{ m}^2$ follows Kiyan's recommendation for correcting the three-dimensional geometry mathematical formula.

Calculator 1 settings:

- Attribute Type: Point Data
- Result Array Name: DragForce
- Expression: $p \cdot \text{Normals}_X$

This computes the drag force at every surface point. OpenFOAM stores pressure as kinematic pressure (p/ρ), so fluid density is already divided out and no additional ρ term is needed.

Calculator 2 settings:

- Attribute Type: Cell Data or Point Data (whichever one has DragForce as a scalar)

- Result Array Name: Cd
- Expression: $2 * \text{DragForce} / (U^2 * D * L)$
- In this case $D = 8$, $L = 40$, $U = 0.5$
- *note that this is not the same as text book formula, since textbook formula focuses on 2D cases whereas this is a 3D case*

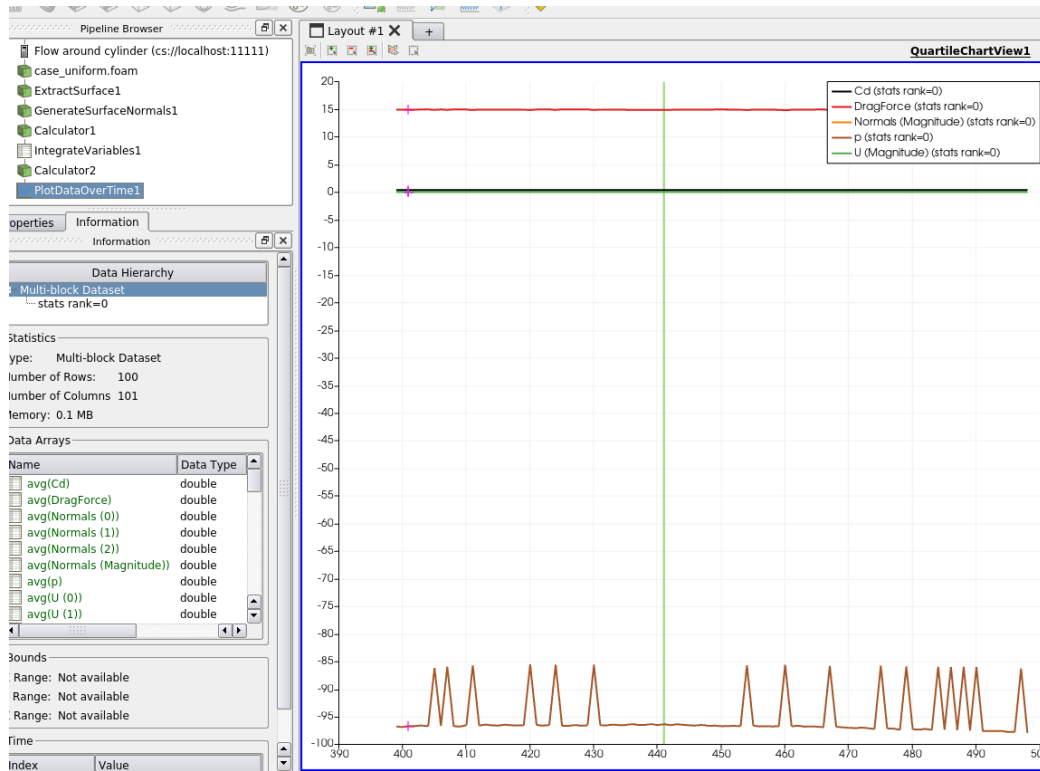


Figure 3.1b — ParaView pipeline and QuartileChartView of Cd, DragForce, and related fields over time.

3.2 Mean Back Pressure Coefficient (Cpb)

Physical meaning: Pressure coefficient at the rear stagnation point (the point on the cylinder surface opposite to the incoming flow). It indicates the strength of the wake suction region.

Formula:

$$C_p(\theta, t) = \frac{p(\theta, t) - p_\infty}{\frac{1}{2} \rho U_\infty^2}$$

Mean value (at the back point, $\theta = 180^\circ$):

$$\overline{C_{pb}} = \frac{1}{T} \int_0^T C_p(\theta = 180^\circ, t) dt$$

From simulation: Extract the pressure at the back point (wake side of cylinder surface) and time-average over cycles.

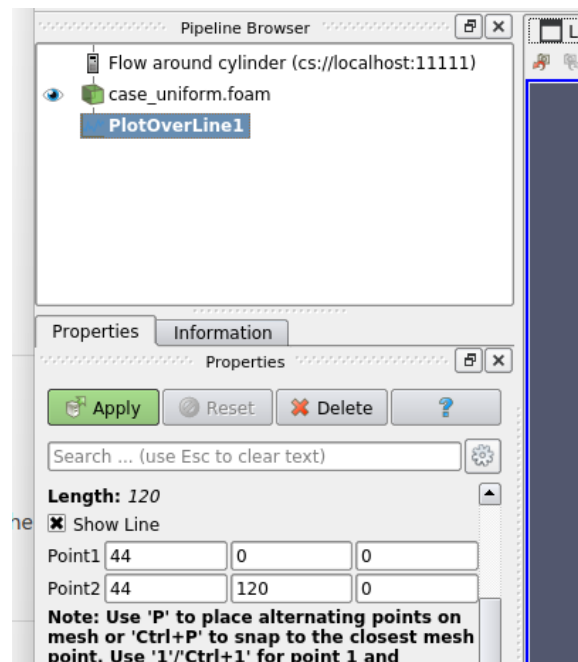
Figure 3.2a — Physical meaning, formula, and mean-value definition of the back pressure coefficient.

Variable	Value
x-location of line probe	44 m (this was gotten from paraview's information tab, X: 36 to 44, Y: 56 to 64, Z: -20 to 20)
y-location at centerline	60 m
Base point (Point ID 500, on cylinder wall where $U_{\text{average}} \approx 0$)	(44, 60, 0)
Time-averaged base pressure at (44, 60, 0)	$-0.05985 \text{ m}^2/\text{s}^2$ (kinematic)
Far field reference pressure, p_{∞} (line ends)	$-0.00245 \text{ m}^2/\text{s}^2$
Freestream velocity, U_{∞}	0.5045 m/s
Dynamic pressure, $0.5U_{\infty}^2$	$0.12727 \text{ m}^2/\text{s}^2$
Time-Averaged C_{pb}	-0.451

Farfield pressure and freestream velocity don't need to be averaged because they don't really change.

The computed time averaged back pressure coefficient of -0.451 falls within the range of approximately -0.4 to -0.6 reported in experimental literature for circular cylinders in the transcritical regime near $Re = 4 \times 10^6$.

PlotOverLine Settings (note the "x" values of the point needs to be 44)



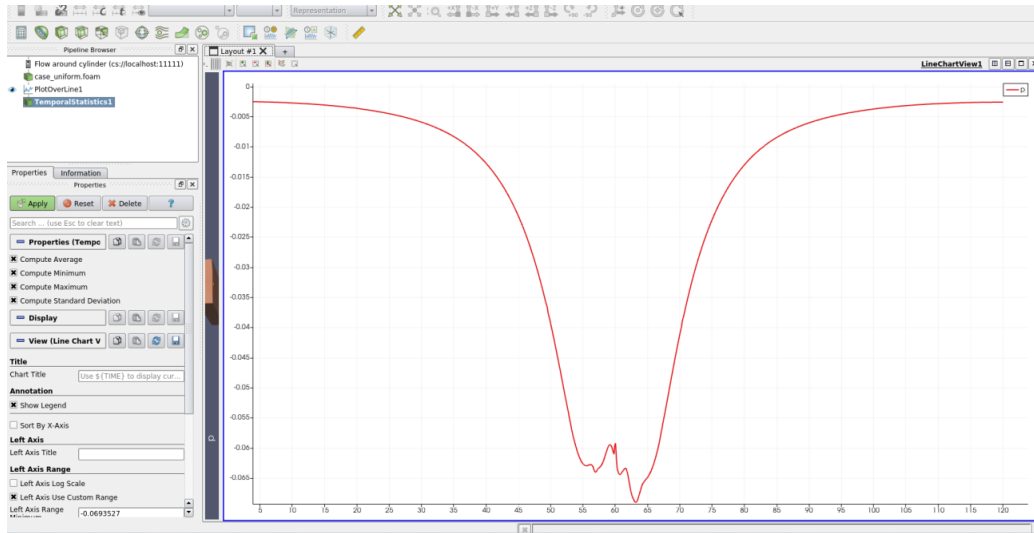


Figure 3.2b — PlotOverLine settings (top) and the resulting pressure-deviation curve along the probe line (bottom).

X-axis shows the y-coordinate at $x=44\text{m}$, y-axis shows how much pressure deviates from freestream (with pressure at freestream = 0)

The curve has three distinct regions:

- Farfield (at both ends $y < 45$ and $y > 70$): The pressure is nearly flat at around -0.002 to $-0.003 \text{ m}^2/\text{s}^2$. This is the undisturbed freestream reference pressure p_∞ . The flow hasn't been significantly affected by the cylinder here
- Approach and recovery ($y = 45 \sim 55$, $y = 65 \sim 75$): pressure drops steeply as the probe enters the cylinder's wake region. On the downstream side it recovers gradually, which is typical of a turbulent wake flow
- Cylinder base region ($y = 55 \sim 65$): The pressure reaches a minimum of about $-0.060 \text{ m}^2/\text{s}^2$. This is the base pressure.

Pressure Plot

3.3 Strouhal Number (St)

Physical meaning: A dimensionless measure of vortex shedding frequency.

Formula:

$$St = \frac{f_s D}{U_\infty}$$

where f_s is the vortex shedding frequency, D the cylinder diameter, and U_∞ the free-stream velocity.

From simulation:

- Monitor a velocity or lift coefficient signal over time.
- Perform FFT or spectral analysis to determine the dominant frequency f_s .
- Normalize by D and U_∞ to get St .

Figure 3.3 — Physical meaning, formula, and extraction method for the Strouhal number.

Variable	Value
Dominant Shedding Frequency f	3/95 Hz
Cylinder Diameter D	8 m
Freestream Velocity U	0.5 m/s
Strouhal Number St	0.505263158

FFT analysis was attempted first using a lift coefficient (Cl) time series. This method requires the forceCoeffs function object, which was not configured in this case. No Cl data was available, so the FFT method could not be used.

Instead, the shedding frequency was found manually from the ParaView animation. One full shedding cycle was counted as one vortex detaching from the top, followed by one from the bottom. Three full cycles occurred over 95 frames (in the animation I extracted 1 frame = 1 second), giving $f = 3/95$ Hz.

$$St = fD/U = (3/95)(8)/(0.5) = 0.5053$$

A shedding period of 3/95 Hz means one full vortex pair sheds about every 32 seconds. This St value shows periodic, organized vortex shedding. It is close to the values reported by Miralles and Koobus (2023) and Yeon et al. (2016) for similar high-Reynolds-number simulations.

3.4 Mean Recirculation Length (Lr/D)

The recirculation length was extracted from a Plot Over Line filter along the wake centerline, identifying the downstream location where mean streamwise velocity U_x returns to zero.

4. Mean Recirculation Length (L_r)

Physical meaning: The distance from the cylinder rear surface to the downstream point where the time-averaged streamwise velocity along the centerline (wake axis) recovers to zero. It characterizes the length of the recirculating wake bubble.

From simulation:

- Compute the time-averaged streamwise velocity field $\bar{u}(x, 0)$ along the wake centerline.
- Find the downstream location where $\bar{u}$ first becomes positive (recovery point).
- The distance between cylinder rear and that point is L_r .

In Paraview:

1. Use the temporal statistics filter to calculate time averaged data (want average U_x)
2. Slice filter at the y normal centered along the wake centerline
3. Slice filter at the z normal in the center ($z = 0$ in my case)
4. Find where $\overline{U_x}$ first becomes positive, either the point where $\overline{U_x} = 0$ or the two points downstream (the negative and the positive point) where the transition happens (I used measurements in the center of the segments)
5. *Skip if found 0 point:* Use linear interpolation to find the estimated 0 point by finding slope and intercept then solving for x when $y(U_x) = 0$
6. Calculate L_r = transition point - backpoint of cylinder
7. For comparison calculate $\frac{L_r}{D}$, divide your L_r by the cylinder diameter

Figure 3.4a — Physical meaning, from-simulation procedure, and ParaView steps for the mean recirculation length.

Variable	Value
x-position of cylinder back face	X = 44 m
Downstream location	X = 53.713 m
L_r	9.713 m (by linear interpolation)
L_r/D (dimensionless recirculation length)	1.214

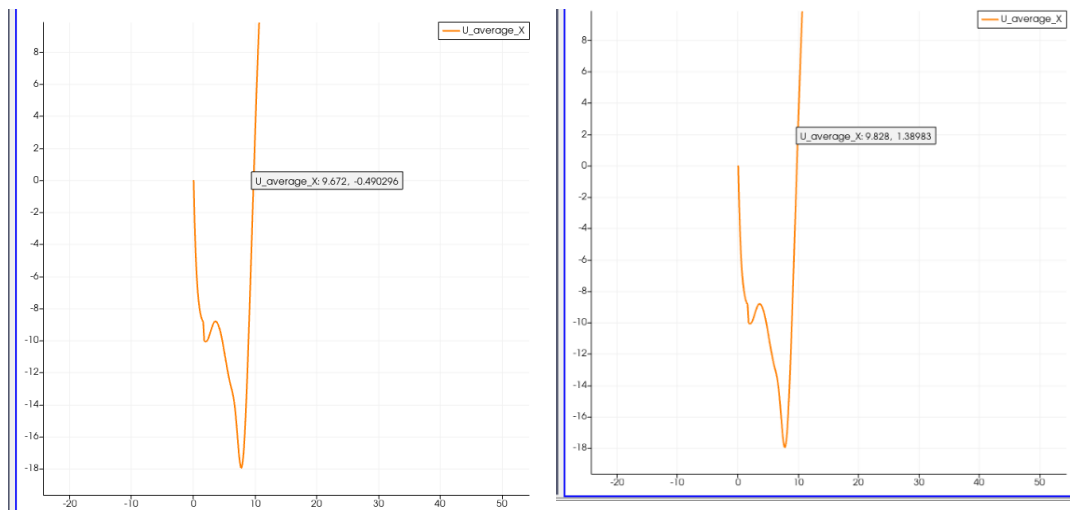
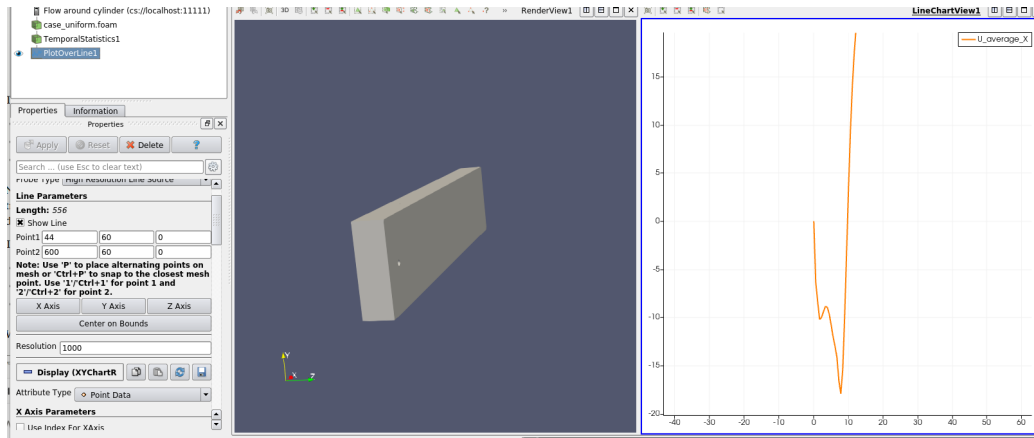


Figure 3.4b — Pipeline, 3D geometry, and $U_{average_X}$ profile along the wake centerline, with the zero-crossing bracketed by point 1 and point 2.

Point 1 (9.828, 1.38983), point 2 (9.672, -0.490296)

X axis represent distance downstream from the cylinder rear face

Y axis shows the time-averaged streamwise velocity $U_{average_X}$

The temporal statistics already solves for the time-averaged value, so the plot overline is sampling from time-averaged field. Therefore, the $L_r = 9.713$ m solved here is the mean recirculation length.

The recirculation length was extracted by first applying Temporal Statistics to compute the time-averaged velocity field, then using a Plot Over Line filter along the wake centerline from (44, 60, 0) to (600, 60, 0), identifying the downstream location where mean streamwise velocity U_x returns to zero. Linear interpolation was applied between the two bracketing points (9.672, -0.490296) and (9.828, 1.38983) to determine the exact zero crossing.

The mean recirculation length of $L_r/D = 1.214$ indicates that the reverse flow region extends approximately 1.2 diameters downstream of the cylinder before the mean streamwise velocity recovers to zero. This relatively compact recirculation bubble is consistent with supercritical flow

behavior at $Re = 4 \times 10^6$, where the turbulent boundary layer delays separation and produces a narrower, shorter wake compared to subcritical regimes. The result falls within the expected range reported in literature for high Reynolds number flow past a circular cylinder, confirming that the DDES simulation is capturing the time-averaged wake structure accurately.

3.5 Mean Separation Angle

The separation angle was identified from the surface pressure or wall shear stress distribution around the cylinder, measured from the front stagnation point. It is conventionally identified as the point where the pressure coefficient stops following the attached-flow trend and deviates (minimum Cp or inflection point on Cp vs Angle graph).

3. Mean Separation Angle ($\bar{\theta}_s$)

Physical meaning: The angular position (measured from the front stagnation point) where the boundary layer separates from the cylinder surface. It reflects the effectiveness of flow attachment and wake formation.

Determination: At a given time, the separation angle is typically where the wall shear stress (or surface velocity tangent to the wall) changes sign.

From simulation:

- Compute the tangential velocity at points on the cylinder surface.
- Locate the angle where it crosses zero (separation point).
- Do this for both sides of the cylinder and average.
- Time-average over cycles to get the mean separation angle.

Figure 3.5a — Physical meaning, determination method, and from-simulation procedure for the mean separation angle.

Variable	Value
Separation Angle (from front stagnation point)	117°
Method Used	Surface Pressure

Method: pipeline and specific filter settings shown

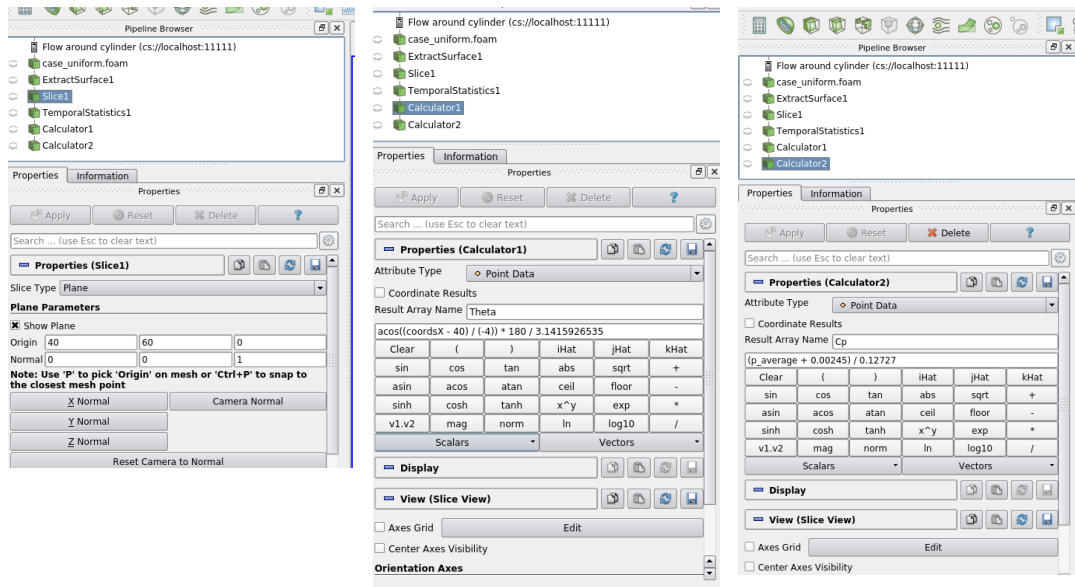


Figure 3.5b — ParaView pipeline: Slice, TemporalStatistics, and the Theta and Cp Calculator settings.

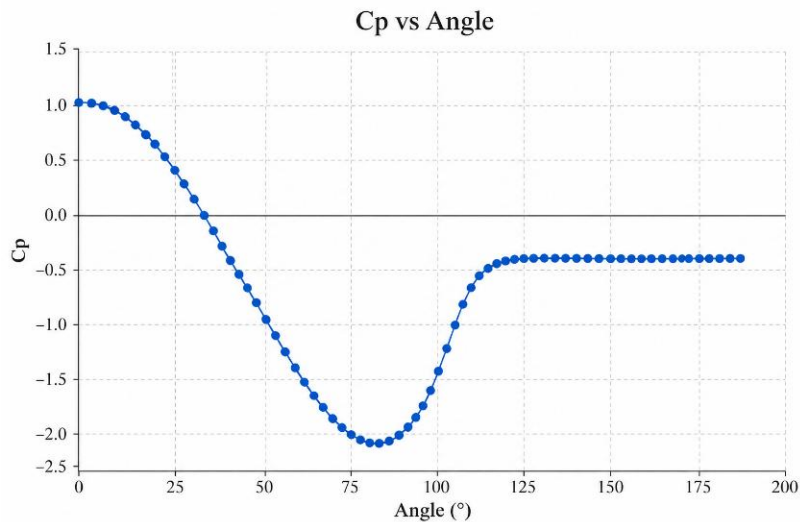


Figure 3.5c — Cp vs Angle plot used to identify the separation angle.

Cp vs Angle Plot

The y axis represents the Cp (note that Cp is called the coefficient of pressure but is infact representing a magnitude instead of “rate of change”)

The y-axis represents angle in degrees.

The mean separation angle was extracted from the time-averaged surface pressure coefficient distribution around the cylinder mid-span cross-section. The pipeline consisted of case.foam → ExtractSurface → Slice (Normal (0,0,1), Origin (40,60,0)) → TemporalStatistics → Calculator (Theta = $\text{acos}((\text{coordsX} - 40)/(-4)) \times 180/\pi$) → Calculator ($\text{Cp} = (\text{p_average} + 0.00245)/0.12727$), with separation identified at the angle where the Cp starts flattening.

I ran a Python script to identify the separation angle from the Cp vs theta data. After averaging the top and bottom halves of the cylinder surface, I found the minimum Cp at 81°. Then, looking only at angles beyond that minimum, I computed the gradient (first derivative) of Cp with respect to theta and found the first angle where the gradient magnitude stayed below 0.005 for at least 5 consecutive points, indicating the curve had flattened into the wake plateau. The resulting separation angle is 117°.

```
import pandas as pd
import numpy as np

df = pd.read_csv("CpVSTheta.csv")
df = df[['Theta', 'Cp']].dropna()

# Average top and bottom halves of cylinder (duplicate Theta values)
df = df.groupby('Theta')['Cp'].mean().reset_index()
df = df.sort_values('Theta').reset_index(drop=True)

# Find minimum Cp
min_idx = df['Cp'].idxmin()
print(f"Minimum Cp = {df.loc[min_idx, 'Cp']:.4f} at Theta = {df.loc[min_idx, 'Theta']:.2f}°")

# Look only after the minimum for the flat wake region
post_min = df.loc[min_idx:].reset_index(drop=True)
dCp_post = np.gradient(post_min['Cp'].values, post_min['Theta'].values)
```

```
threshold = 0.005
flat = np.abs(dCp_post) < threshold

for i in range(len(flat) - 4):
    if all(flat[i:i+5]):
        print(f"Separation angle = {post_min.loc[i, 'Theta']:.2f}° (Cp = {post_min.loc[i,
'Cp']:.4f})")
        break
```

4. Flow Visualizations

4.1 Wake Vorticity Structure

To compute vorticity, apply a Gradient filter on the case with U selected as the scalar array. In the Properties panel, check Compute Vorticity and click Apply. Change the coloring to Vorticity Z component in the toolbar. Since the automatic color range does not show the wake structures, I manually rescaled the range to -0.05 to 0.05 to make the alternating vortex patterns visible.

The vorticity Z visualization shows a highly disorganized near wake immediately behind the cylinder, dominated by small scale turbulent structures characteristic of fully turbulent separation at $Re = 4 \times 10^6$. Coherent large scale vortical structures emerge further downstream but break down rapidly, reflecting the three dimensional and aperiodic nature of high Reynolds number wake dynamics. This contrasts with the clean periodic von Kármán street observed at subcritical Reynolds numbers, where laminar separation produces well organized alternating vortices.

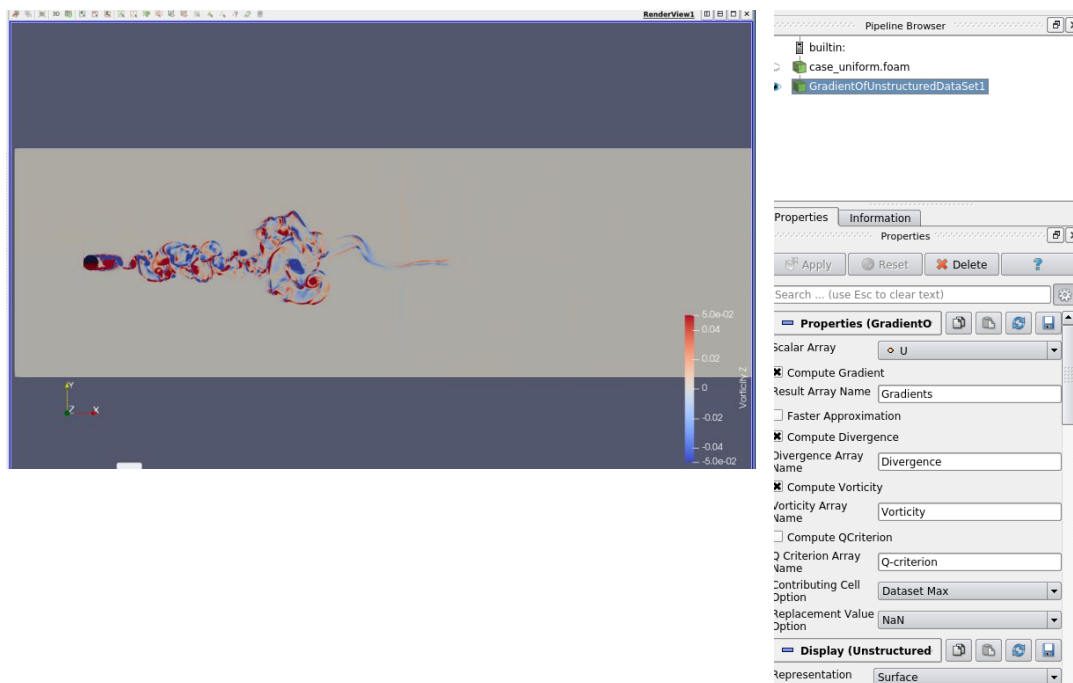


Figure 4.1 — Vorticity Z render (left) and the Gradient filter properties used to compute it (right).

4.2 Q-Criterion Isosurface

In the Gradient filter, check Compute QCriterion, then apply a Contour filter with Q-criterion selected and the isosurface value set to a small number like 0.001 to capture only the strongest vortex cores without too much noise. The Q-criterion isosurface colored by U Z reveals complex three-dimensional vortex tube structures in the wake, with tight compact cores near the cylinder that grow and twist further downstream. The expanding and increasingly chaotic structures reflect the highly turbulent nature of the wake at $Re = 4 \times 10^6$.

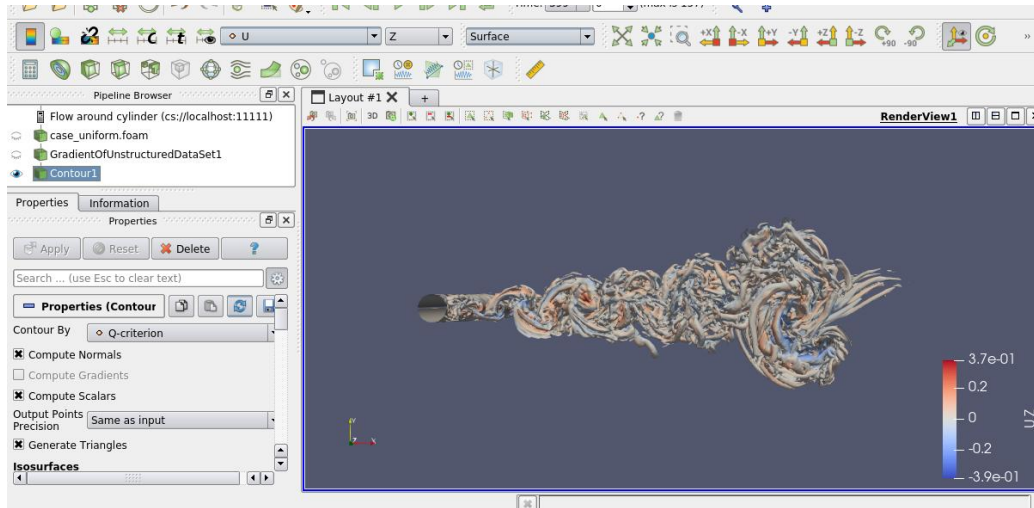


Figure 4.2 — *Q-criterion isosurface, colored by U Z, revealing three-dimensional vortex tube structures.*

Pipeline and model

5. Comparison with Published Literature

5.1 Papers Reviewed

Reference	Key Data Reported	Re closest to 4×10^6
Miralles and Koobus (2023)	Cd, Cpb, St, separation angle	1×10^6
Yeon et al. (2016)	Cd, St, separation angle	7.57×10^5
Roshko (1961)	Cd, St	1×10^6 to 1×10^7
Catalano et al. (2003)	Cd, Cpb, St, Lr	1×10^6
Rodriguez et al. (2015)	Cd, Cpb, Lr	8.5×10^5

5.2 Data Extraction Method

Values for Catalano et al. (2003) were read directly from Table 1, and values for Miralles and Koobus (2023), Yeon et al. (2016), and Rodríguez et al. (2015) were read directly from their respective summary tables. For Roshko (1961), the values of $Cd = 0.70$ and $St = 0.27$ were stated explicitly in the text on pages 351–352, and Cpb was read from the text alongside Figure 3. No digitization was required for any of the five papers as all key values were available in tabular or explicit text form.

Where a paper's Reynolds number range did not extend to $Re = 4 \times 10^6$, the closest available data point was used. For Roshko (1961), the reported values apply for $Re > 3.5 \times 10^6$ which directly covers the target $Re = 4 \times 10^6$. For all other papers, the highest reported Re was used, which ranged from $Re = 7.57 \times 10^5$ (Yeon et al.) to $Re = 10^6$ (Catalano et al. and Miralles and Koobus). These differences in Re are noted in Section 5.4 when interpreting discrepancies between simulation results and literature values.

5.3 Quantitative Comparison Table

Quantity	This Simulation	Miralles & Koobus, 2023	Yeon et al. 2016	Roshko 1961	Catalano et al. 2003	Rodriguez et al. 2015
Re	4×10^6	1×10^6	7.57×10^5	1×10^6 to 1×10^7	1×10^6	8.5×10^5
Mean Cd	0.3742	0.289	0.23	0.70	0.31	0.218
Mean Cpb	-0.451	-0.25	-0.24	-0.86	-0.32	-0.207
St	0.505	0.50	0.46	0.27	0.35	—
Separation Angle	117°	128°	135°	—	110°	—
Recirculation Lr/D	1.214	—	—	—	1.04	—

Appendix

A. Raw Extracted Literature Data

A.1 Miralles and Koobus (2023)

Source of Data:

Table 4 — Bulk coefficients of the flow around a circular cylinder at Reynolds numbers 7.2×10^5 and 10^6 (super-critical regime).

Selected Dataset:

k- ϵ - γ /DVMS model at $Re = 1 \times 10^6$.

Note:

Closest available Re to this simulation is 10^6 , which is $4\times$ lower than the target $Re = 4\times 10^6$ of the present work.

Table 4
Bulk coefficients of the flow around a circular cylinder at Reynolds numbers 7.2×10^5 and 10^6 (super-critical regime). Same symbols as in Table 2.

	$\bar{C}_d$	$ \bar{C}_L $	C'_l	$-\bar{C}_{ps}$	θ_{sep}	St
Present simulation $Re = 7.2 \times 10^5$						
k- ϵ /DVMS	0.548	0.01	0.31	0.50	110	0.32
k- ϵ - γ /DVMS	0.246	0.00	0.01	0.22	125	0.43
Simulation						
LES of Lehmkühl et al. (2014), Rodriguez et al. (2015)	0.21	0.02	0.07	0.22	—	0.45
Experiments						
Schewe (1983)	0.22	—	0.02	—	—	0.47
Gölling (2006)	0.21	—	—	—	—	0.49
Roshko (1961)	0.27	—	—	0.24	—	—
Present simulation $Re = 10^6$						
k- ϵ /DVMS	0.536	0.03	0.30	0.51	110	0.34
k- ϵ - γ /DVMS	0.289	0.03	0.04	0.25	128	0.50
Simulation						
LES of Kim and Mohan (2005)	0.27	—	0.12	0.28	108	—
LES of Catalano et al. (2003)	0.31	—	—	0.32	—	0.35
Experiments						
Shih et al. (1993)	0.24	—	—	0.33	—	—
Schewe (1983)	0.22	—	0.02	—	—	0.44
Gölling (2006)	0.22	—	—	—	130	0.12/0.47
Zdravkovich (1997)	0.2–0.4	—	0.1–0.15	0.2–0.34	—	0.18/0.50
Roshko (1961)	0.29	—	—	0.34	—	—

Figure B1. Table 4 from Miralles and Koobus (2023) showing bulk coefficients in the super-critical regime.

Intermittency (γ) is a variable between 0 and 1 that represents the fraction of time the flow is turbulent at a given point. It allows the model to capture the gradual transition from laminar to turbulent rather than assuming the flow is either fully laminar or fully turbulent everywhere.

k- ϵ - γ /DVMS adds the intermittency equation, so it can model where and how the boundary layer transitions from laminar to turbulent. Since DDES simulation (ours), also captures transition effects so we use this data.

A.2 Yeon et al. (2016)

Source of Data:

Table 3 — Summary of integral and local variables in the super-critical regime.

Selected Dataset:

Dynamic SGS model at $Re = 7.57 \times 10^5$.

Note:

$Re = 7.57 \times 10^5$ is the highest Re reported in this paper, approximately $5\times$ lower than the target $Re = 4 \times 10^6$ of the present work. D and S refer to dynamic and static SGS model variants respectively.

Table 3
Summary of integral and local variables with experimental data.

	Re		C_D	θ_{LS}	θ_{TS}	$-C_{pb}$	C_L^{RMS}	St	λ/D
Sub-critical	6.31×10^4	D	1.20	81	–	1.30	0.55	0.20	3.5
		S	1.37	82	–	1.46	0.60	0.19	3.3
	1.26×10^5	D	1.16	75	–	1.32	0.56	0.20	3.0
		S	1.37	81	–	1.64	0.62	0.20	2.9
Critical	2.52×10^5	D	0.92	94	–	1.00	0.17	0.19	2.4
		S	0.56	89	–	0.44	0.12	0.19	0.8
Super-critical	4.15×10^5	D	0.36	–	136	0.24	0.021	0.47	1.2
		S	0.27	95	113	0.30	0.064	0.41	–
	5.06×10^5	D	0.29	–	139	0.24	0.023	0.47	1.1
		S	0.25	96	113	0.25	0.061	0.41	–
	7.57×10^5	D	0.23	–	135	0.24	0.027	0.46	1.1
		S	0.21	94	112	0.34	0.042	0.34	1.3

Figure B2. Table 3 from Yeon et al. (2016) showing super-critical flow parameters.

A.3 Roshko (1961)

Source of Data:

Figure 2 (C_d vs Re), Figure 3 (C_{pb} vs Re), Figure 5 (St vs Re), and accompanying discussion on pages 351–352.

Table B3. Extracted values from Roshko (1961).

Quantity	Value	Re	How obtained
C_d	~ 0.70	$Re > 3.5 \times 10^6$	Stated explicitly in text, page 352
C_{pb}	-0.86	$Re = 2 \times 10^5$ to 3.5×10^6	Stated explicitly in text, page 352
St	0.27	$Re > 3.5 \times 10^6$	Stated explicitly in text, page 351

Note:

$Re = 4 \times 10^6$ falls in Roshko's upper transition range where C_d has just reached its plateau of ~ 0.70 . C_{pb} value of -0.86 applies to the transition range just below $Re = 3.5 \times 10^6$ and may differ slightly at $Re = 4 \times 10^6$. $St = 0.27$ applies for $Re > 3.5 \times 10^6$ which directly covers your simulation Re .

A.4 Cantanalo et al. (2003)

Source of Data:

Table 1 — Drag coefficient, base pressure coefficient, Strouhal number, and recirculation length at $Re = 1 \times 10^6$.

Selected Dataset:

LES row.

Note:

$Re = 1 \times 10^6$ is $4\times$ lower than the target $Re = 4 \times 10^6$ of the present work. This is the only paper reviewed that explicitly reports L_r/D at high Re , making it the primary recirculation length reference despite the Re difference.

Table 1 Drag, base pressure coefficient, Strouhal number, and recirculation length for the flow around a circular cylinder at $Re_D = 1 \times 10^6$				
	C_D	$-C_{P_{base}}$	St	L_r/D
LES	0.31	0.32	0.35	1.04
RANS	0.39	0.33	—	—
URANS	0.40	0.41	0.31	1.37
Exp. (Shih et al., 1993)	0.24	0.33	0.22	—
Exp. (Others, see Zdravkovich, 1997)	0.17–0.40	—	0.18–0.50	—

Figure B4. Table 1 from Catalano et al. (2003) showing high-Reynolds-number wake properties.

A.5 Rodríguez et al. (2015)

Source of Data:

Table 2 — Flow parameters.

Table 3 — Critical parameters of the near wake.

Selected Dataset:

Re = 8.5×10^5 row.

Notes:

Re = 8.5×10^5 is the highest Re in this paper, approximately 5× lower than the target Re = 4×10^6 . An interesting trend visible in Table 3 is that L_r/D does not decrease monotonically with Re — it peaks at Re = 5.3×10^5 ($L_r/D = 0.727$) then stabilizes around 0.597 at Re = 8.5×10^5 , suggesting the recirculation length plateaus in the supercritical regime.

Table 2
Flow parameters. Drag coefficient C_D ; fluctuating drag $C_{D,rms}$; lift coefficient C_L ; fluctuating lift $C_{L,rms}$; cylinder base pressure $-C_{p,base}$.

Re	C_D	$C_{D,rms}$	C_L	$C_{L,rms}$	$-C_{p,base}$
2.5×10^5	0.833	0.095	-0.903	0.486	0.99
3.8×10^5	0.481	0.061	0.245	0.217	0.48
5.3×10^5	0.296	0.011	0.0614	0.0705	0.305
6.5×10^5	0.232	0.0081	0.027	0.0755	0.23
7.2×10^5	0.213	0.0094	0.016	0.0752	0.224
8.5×10^5	0.218	0.0092	0.007	0.070	0.207

Table 3
Critical parameters of the near wake. L_r , non-dimensional length of the recirculation zone; L_f , non-dimensional vortex formation length; $L_{u,min}$, u_{min} non-dimensional location and magnitude of the minimum stream-wise velocity u/U_{ref} in the wake centreline, respectively; x_c and y_c , non-dimensional stream-wise and cross-stream coordinates of the location of the mean wake vortex centre.

Re	L_r	L_f	$L_{u,min}$	u_{min}	Vortex centre	
					x_c	y_c
2.5×10^5	0.559	1.105	0.757	-0.194	0.711/0.707	0.308/-0.168
3.8×10^5	0.641	1.107	0.778	-0.205	0.73/0.728	0.199/-0.2
5.3×10^5	0.727	1.183	0.885	-0.253	0.831/0.825	0.153/-0.154
6.5×10^5	0.596	1.044	0.86	-0.262	0.807/0.822	0.132/-0.129
7.2×10^5	0.569	1.062	0.854	-0.254	0.813/0.807	0.125/-0.128
8.5×10^5	0.597	1.091	0.871	-0.23	0.819/0.820	0.135/-0.134

Figure B5. Tables 2 and 3 from Rodríguez et al. (2015) showing wake parameters and recirculation length trends.

B. References

- Catalano, P., Wang, M., Iaccarino, G., & Moin, P. (2003). Numerical simulation of the flow around a circular cylinder at high Reynolds numbers. *International Journal of Heat and Fluid Flow*, 24, 463–469.
- Rodríguez, I., Lehmkuhl, O., Chiva, J., Borrell, R., & Oliva, A. (2015). On the flow past a circular cylinder from critical to super-critical Reynolds numbers: Wake topology and vortex shedding. *International Journal of Heat and Fluid Flow*, 55, 91–103.
- Roshko, A. (1961). Experiments on the flow past a circular cylinder at very high Reynolds number. *Journal of Fluid Mechanics*, 10, 345–356.
- Miralles, F., & Koobus, B. (2023). Simulation of the flow past a circular cylinder from sub-critical to super-critical Reynolds numbers using an intermittency-based hybrid model. *Journal of Fluids and Structures*, 123, 104008.
- Yeon, S. M., Yang, J., & Stern, F. (2016). Large-eddy simulation of the flow past a circular cylinder at sub- to super-critical Reynolds numbers. *Journal of Fluids and Structures*, 61, 93–108.